TEK4090 – Introduction to Modern Control
Assignment 1

Due: September 18 2025, 13:15

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1. (10 points) Suppose that $v_1, \ldots, v_m$ is linearly independent in V, and that $w \in V$. Show that

$$v_1, \ldots, v_m, w \text{ is linearly independent} \iff w \notin \operatorname{span}(v_1, \ldots, v_m). \tag{1}$$

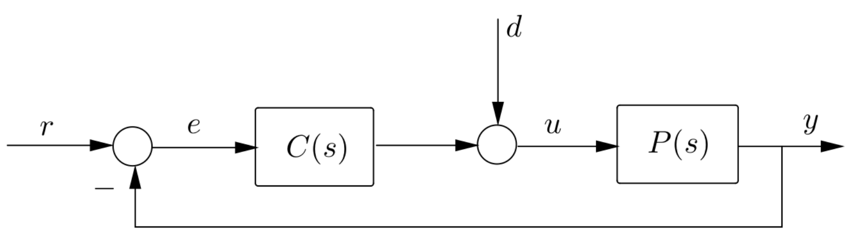
2. (10 points) Let $A, B \in \mathbb{R}^{n\times n}$, and define the *trace* of A as $\operatorname{tr}(A) = \sum_{i=1}^{n} A_{ii}$. Show that

$$\frac{d\operatorname{tr}(AB)}{dA} = B^T. \tag{2}$$

Hint: Find $\frac{d\operatorname{tr}(AB)}{dA_{ij}}$.

3. (15 points) Suppose that $T : V \mapsto V$ is a linear map, and suppose that $S : V \mapsto V$ is an invertible linear map.
   (a) Prove that T and $S^{-1}TS$ have the same eigenvalues
   (b) Describe the relationship between the eigenvectors of T and the eigenvectors of $S^{-1}TS$.

4. (30 points) Consider the feedback block diagram:



   (a) Find the transfer function from d to y.
   (b) Suppose the plant has transfer function

$$P(s) = \frac{(s+1)}{s^2 - 4s + 13} \tag{3}$$

   What are the poles of $P(s)$? What can you infer about the stability of the (open-loop) plant $P(s)$?

(c) Suppose $C(s)$ is given by

$$C(s) = \frac{1}{s} \tag{4}$$

Does $C(s)$ stabilize the feedback system?

(d) Now, suppose the plant and controller are given by

$$P(s) = \frac{10}{(s+1)},\ C(s) = \frac{K}{s}. \tag{5}$$

Find the values of K for which the feedback loop is stable.

(e) Fix a value of K such that the system in part (d) is stable, and suppose the system is subject to a disturbance $D(s) = 100$. What will the final value of y be?

5. (10 points) Consider the standard feedback system with

$$C(s) = K,\ P(s) = \frac{s^2+1}{(s^2+2)(s-2)} \tag{6}$$

(a) Sketch the Nyquist plot of $P(s)$. Include correct asymptotes to infinity and to the origin.

(b) Indicate the number of encirclements (and their orientations) of all possible critical points.

(c) For what range of K is the feedback system stable?

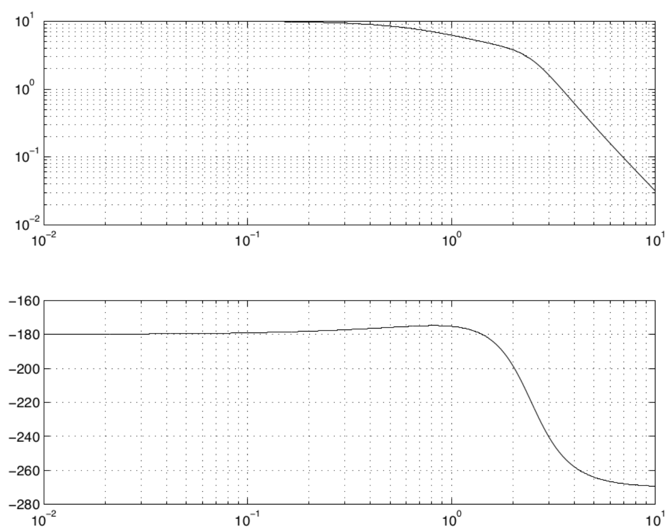
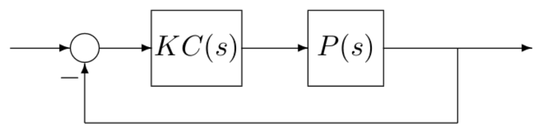


Fig. 1: Bode Plot for Question (6)

6. (15 points) Consider the standard feedback system with $C(s) = K$. The Bode plot of $P(s)$ is given in Figure 1 (magnitude in absolute units, not dB; phase in degrees). The phase starts at $-180°$ and ends at $-270°$. You are also given that $P(s)$ has exactly one pole in the right half-plane. For what range of gains K is the feedback system stable?

7. (10 points) Consider the feedback system:



with

$$P(s) = \frac{s+2}{s^2+2}, \; C(s) = \frac{1}{s}. \tag{7}$$

Sketch the Nyquist plot of PC. How many encirclements are required of the critical point for feedback stability? Determine the range of real gains K for stability of the feedback system.

8. (10 points) Repeat the previous question with,

$$P(s) = \frac{s^2+1}{(s+1)(s^2+s+1)}, \; C(s) = 1. \tag{8}$$